Journal of Computational Physics 231 (2012) 6419–6437

Contents lists available at SciVerse ScienceDirect

Journal of Computational Physics

journal homepage: www.elsevier.com/locate/jcp

Anisotropic mesh adaptation on Lagrangian Coherent Structures

Philippe Miron [a], Jérôme Vétel [a], André Garon [a,*], Michel Delfour [b,a], Mouhammad El Hassan [a]

[a] LADYF – Laboratoire de Dynamique des Fluides, École Polytechnique de Montréal, C.P. 6079, succ. Centre-ville, Montréal (Québec), Canada H3C 3A7
[b] Centre de recherches mathématiques & Département de mathématiques et de statistiques, Université de Montréal, C.P. 6128, succ. Centre-ville, Montréal (Québec), Canada H3C 3J7

ARTICLE INFO

Article history:
Received 25 August 2011
Received in revised form 28 May 2012
Accepted 6 June 2012
Available online 20 June 2012

Keywords:
Finite time Lyapunov exponent
Lagrangian Coherent Structure
Continuous metric
Anisotropic mesh adaptation
Error estimation
Particle image velocimetry

ABSTRACT

The finite-time Lyapunov exponent (FTLE) is extensively used as a criterion to reveal fluid flow structures, including unsteady separation/attachment surfaces and vortices, in laminar and turbulent flows. However, for large and complex problems, flow structure identification demands computational methodologies that are more accurate and effective. With this objective in mind, we propose a new set of ordinary differential equations to compute the flow map, along with its first (gradient) and second order (Hessian) spatial derivatives. We show empirically that the gradient of the flow map computed in this way improves the pointwise accuracy of the FTLE field. Furthermore, the Hessian allows for simple interpolation error estimation of the flow map, and the construction of a continuous optimal and multiscale L^p metric. The Lagrangian particles, or nodes, are then iteratively adapted on the flow structures revealed by this metric. Typically, the L^1 norm provides meshes best suited to capturing small scale structures, while the L^∞ norm provides meshes optimized to capture large scale structures. This means that the mesh density near large scale structures will be greater with the L^∞ norm than with the L^1 norm for the same mesh complexity, which is why we chose this technique for this paper. We use it to optimize the mesh in the vicinity of *LCS*. It is found that Lagrangian Coherent Structures are best revealed with the minimum number of vertices with the L^∞ metric.

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1. Introduction

It is common knowledge that a two-dimensional (2D) steady flow structure, such as a recirculation zone or a stagnation/separation line, is easily identified with the use of streamlines. However, the analysis of three-dimensional (3D) flows is much more difficult, and pathlines or streaklines often fail to reveal the complexity and interactions of flow structures in a rigorous frame independent setting. Haller's [1–3] response to this issue stemmed from the theory of dynamic systems applied to the study of particles passively transported by the flow. In this theory, the notion of Lyapunov stability appears, and results in the construction of the finite time Lyapunov exponent (FTLE). He showed that the ridges of this field define material lines known as Lagrangian Coherent Structures (*LCS*), which are used to accurately identify vortices, as well as stagnation/separation and attachment lines and surfaces.

The article of Shadden et al. [4] describes a methodology for calculating the Lyapunov exponent in $\mathbb{R}^3$ on a regular Cartesian grid aligned with the coordinate system. This grid is superposed on the region of interest through which a fluid is flowing. Where the grid lines intersect determines the initial coordinates of the particles transported by the flow from a time t_0 to

* Corresponding author.
E-mail addresses: philippe.miron@polymtl.ca (P. Miron), jerome.vetel@polymtl.ca (J. Vétel), andre.garon@polymtl.ca (A. Garon), delfour@crm.umontreal.ca (M. Delfour).

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http://dx.doi.org/10.1016/j.jcp.2012.06.015

a time $t_0 + T$. This computation is performed by the numerical integration of a system of ordinary differential equations using the 4th order Runge–Kutta–Fehlberg method. The position of the particles at time $t_0 + T$ determines the displacement field or flow map, the values of which are known at the grid nodes. The use of a Cartesian grid facilitates calculation of the components of the Cauchy-Green tensor, with the help of approximations, by 2nd order finite differences, of the derivatives of that field. The largest eigenvalue of this tensor is then used to construct the Lyapunov exponent at each grid node, that is, a Lyapunov exponent field in finite time (a FTLE field). This procedure is performed for a series of increasing values of t_0, in order to follow the evolution over time of the FTLE field.

This approach is a very simple one from the algorithmic point of view, if the velocities are known at the nodes of a Cartesian (measurement) grid aligned on the calculation grid. In this case, the interpolation of the calculation grid velocities on the measurement grid is not costly. To improve the quality of the velocity interpolation on the measurement grid, the method of Lieken and Marsden [5] can be used, since it constructs a class C^2 approximation. However, to obtain well-defined Lagrangian structures, the calculation grid must be much more dense than the measurement grid. Without an objective refinement criterion for the detection of coherent structures, the number of intervals in all directions on the calculation grid is systematically and repeatedly doubled, until a FTLE field is achieved that is qualitatively acceptable.

A method for distributing the particles within the calculation grid more effectively was proposed by Garth et al. [6], and involves the calculation of Lyapunov exponents on hierarchical Cartesian grids. For 2D meshes, a quadtree is used to refine the initial regular grid mesh into a regular sub grid that is more dense near flow structures. The refinement is controlled with the help of the Lyapunov exponent interpolation error, which is evaluated at the grid nodes. An element is subdivided into four sub elements when the interpolation error is greater than a predetermined value, but is unchanged if the error is less than this target value. This procedure is applied recursively until the subdivision process is stopped. It is robust, and makes it possible for the velocities to be quickly localized and interpolated for calculation of the Lyapunov exponent. For 3D meshes, this approach is based on the octree data structure.

In this approach, unlike that of Shadden et al. [4], the particles are no longer distributed uniformly, but are concentrated in regions where the gradients of the FTLE field are the largest. With the same number of nodes, it can more effectively capture coherent flow structures, and do so with little added cost generated by error analysis and the subdivision of elements into sub elements. However, the approach as presented does not allow elements to be removed from an initial grid that is too densely populated in some flow regions. This is a limitation that may hamper generalization of their methodology to time-dependent flows.

In Lekien and Ross [7], calculation of the Lyapunov exponents is generalized to differentiable manifolds. Their method is based on the use of unstructured calculation grids adapted to increase the accuracy of coherent structures. At each node, the gradients of the displacement fields are determined with the help of a reconstruction method used in the finite element and finite volume methods [8–11]. The gradients are obtained at a node by a weighted sum of the gradients calculated on the adjacent elements sharing this node. No information is provided on the accuracy of the reconstruction, in particular for the nodes at the edges of the calculation domain, where the accuracy of the reconstruction is known to be generally lower. In their procedure, an element (triangle) is uniformly subdivided into three sub elements, if the value of the field at one of its nodes is greater than a value determined in an ad hoc fashion. This step is performed recursively until a satisfactory qualitatively calculated field is obtained. The approach has the benefit of being extremely robust. The authors show, with the help of a 2D example inspired by Rayleigh-Bénard convection cells (see Section 5 for the description of this flow field), the advantage of combining the isotropic adaptation method with gradient reconstruction. The approach requires a substantially reduced number of nodes for calculating the Lyapunov exponents, compared to previous regular or embedded hierarchical Cartesian grids.

Obtaining a calculation grid that is optimally adapted to *LCS* requires that we move away from hierarchical Cartesian grids and isotropic unstructured meshes. An anisotropic mesh offers maximum flexibility and the potential to significantly reduce the number of nodes necessary for the capture of Lagrangian structures. In addition, elements constructed in this way are deformed to enable them to adapt more effectively to the gradients of the solution. Control of the distribution of calculation nodes is combined with an error estimation method and a continuous metric [12–14] that allows the elements to be stretched and aligned in the calculation domain.

In recent years, the adaptation of anisotropic unstructured grids [15–19] has proved to be an effective way to improve the accuracy of numerical simulations, as well as to capture anisotropic behaviors and phenomena (shock waves, boundary layers, etc.). With this technique, it is possible to significantly reduce the number of grid nodes, while at the same time minimizing the interpolation error on the field. This in turn reduces the memory space and the calculation time required for the solution. The technique is based on the use of triangular (or tetrahedral) elements, which are stretched and aligned in accordance with the gradients of the solution, in order to minimize the interpolation error [20–22].

The most recent advancements in the area of anisotropic meshes have come from the work of Loseille [23] and Alauzet [24]. In particular, Loseille [23] has constructed an original metric as a function of the number of nodes required to approximate the function by finite elements. Even though the applications are in the field of computational fluid dynamics (CFD), the approach can be generally applied to controlling the error of a numerical or vectorial function which is sufficiently regular that the Hessian matrix exists at every point in the calculation grid.

There are various approaches to anisotropic mesh adaptation. In this paper, we use the multiscale continuous metric concept, introduced by [12,14,13], which is suitable for the detection of structures of different sizes. Hence, following the methodology developed by Loseille [23], we define an optimal continuous metric that minimizes the interpolation error

in the L^p norm and read into MeshAdapt [25] which is a commercially available mesh adaptation software. This metric was constructed from the Hessian matrix of the flow map, and not from the Hessian matrix of the FTLE field since that would required more derivatives reconstruction steps. Using analytical examples and measurement data, we demonstrate the capacity and the limitations of this approach to detect and capture *LCS*.

The paper is organized as follows. We recall in Section 2 the main concepts underlying the definition and classical construction of the FTLE. This is followed in Section 3 by the introduction of an extended set of equations that yield its gradient and Hessian concurrently with the flow map. The solutions of these equations are used in the computation of the FTLE field and in the computation of the interpolation error of the flow map in the L^p norm. It is shown in Section 4 that the Hessian of the flow map can also be used to build an optimal anisotropic mesh that minimizes the flow map interpolation error. The optimal Hessian-based metric concept, developed by Loseille, provides the necessary framework for the computation of a continuous metric for the purpose of mesh adaptation. To study how these different concepts can be used effectively together, we present two examples based on the analytical velocity field in Section 5. This is followed in Section 6 and Section 7 by the application of the methodology to the analysis of measured velocity fields acquired with particle image velocimetry. Finally, in Section 8, we provide an assessment of the methodology and the path to follow to further its development.

2. Finite time Lyapunov exponent (FTLE)

FTLE theory has been thoroughly described by Haller [1–3] and Shadden [4] in their seminal papers, and so we recall in this section only the basic definitions and results that lead to the construction of the FTLE.

The original application of the FTLE is the study of fluid flow structures in an open subset U of $\mathbb{R}^n$. For all practical applications, we consider a subset $D \subset U$. This subset could represent either the computational domain, where the Navier–Stokes equations are solved, or the domain where time-resolved measurements of velocity fields are carried out. The following hypotheses and definitions are assumed to apply:

Hypothesis 2.1. *The fluid domain D is open, bounded, and simply connected.* □

Let $\mathbf{u}(x,t)$ be a velocity field that is a function of the space variable $x \in D$ and the time variable t. Assume that $\mathbf{u}$ verifies the following assumptions.

Hypothesis 2.2. *The field $\mathbf{u}(x,t)$ is C^2 with respect to $x \in U$ (that is, $C^2(U)$ and hence $C^2(D)$).* □

Hypothesis 2.3. *There exist t_0 and $\tau > 0$ such that $\mathbf{u}(x,t)$ is C^0 with respect to the time t in the interval $]t_0-\tau, t_0+\tau[$, and that all the pathlines starting from a point x_0 in U at time t_0 do not exist U at any times t in $]t_0-\tau, t_0+\tau[$.* □

This hypothesis allows forward or backward integration in time from t_0 to t_0+T for positive or negative integration time T such that $|T| \leqslant \tau$.

Definition 2.1. Let $\mathbf{u}(\mathbf{x},t)$ be a vector field that satisfies Hypotheses (2.1-2.3) and let $\boldsymbol{\varphi}(t) = \boldsymbol{\varphi}(t;\mathbf{x}_0,t_0)$ be the solution at time t of the differential equation

$$\frac{d\boldsymbol{\varphi}}{dt}(t) = \mathbf{u}(\boldsymbol{\varphi}(t),t) \quad \text{on} \quad]t_0-\tau, t_0+\tau[, \tag{1}$$

$$\boldsymbol{\varphi}(t_0) = \mathbf{x}_0$$

for all initial points $\mathbf{x}_0 \in D$ at t_0. □

Hence $\boldsymbol{\varphi}(t;\mathbf{x}_0,t_0)$ defines the path of the fluid particles in the domain D at time t_0 that remains in U (by Hypothesis 2.2) for all times t in $]t_0-\tau, t_0+\tau[$, or, equally, the deformation of the fluid domain D, from the point of view of Lagrangian kinematics, as it evolves from t_0 to t_0+T remains in U. Lastly, the flow map

$$\boldsymbol{\varphi}_{t_0}^{T}(\mathbf{x}_0) = \boldsymbol{\varphi}(t_0+T;\mathbf{x}_0,t_0) \tag{2}$$

simply corresponds to the final deformation at time t_0+T of the fluid domain as a function of its initial state. The flow map is $C^2(D)$. It is a map $\boldsymbol{\varphi}_{t_0}^{T} : D \to U$ of class C^2.

In continuum mechanics, a change in configuration of D, from the Lagrangian kinematics point of view, results in a displacement measured by the flow map (Eq. (2)).

Definition 2.2. The Cauchy-Green tensor

$$\mathcal{S}(\mathbf{x}_0) = \left(\nabla\boldsymbol{\varphi}_{t_0}^{T}(\mathbf{x}_0)\right)^{t} \cdot \nabla\boldsymbol{\varphi}_{t_0}^{T}(\mathbf{x}_0) \tag{3}$$

is a rotation independent deformation tensor. In Cartesian coordinates, it is written as

$$\mathcal{S}(\mathbf{x}_0) = \sum_{ij} \mathcal{S}_{ij}(\mathbf{x}_0) e_i \otimes e_j$$

$$\mathcal{S}_{ij}(\mathbf{x}_0) = \sum_{k=1}^{n} \left.\frac{\partial \varphi_k}{\partial x_i}\right|_{\mathbf{x}_0} \left.\frac{\partial \varphi_k}{\partial x_j}\right|_{\mathbf{x}_0}. \tag{4}$$

It is a rank two symmetric tensor with real and positive eigenvalues. The largest and the smallest eigenvalues of $\mathcal{S}$ are respectively $\lambda_{\max}(\mathbf{x}_0)$ and $\lambda_{\min}(\mathbf{x}_0)$. □

Note that t_0 and T have been dropped to avoid notation clutter, since these values are implicit by the context. The Cauchy-Green tensor components are $C^1(D)$.

This ordinary differential equation (Eq. (1)) is the classical Lagrangian construction of the flow map. Simple and efficient high order Runge–Kutta integration scheme suffices to compute the flow map with known velocity field, as it is within the scope of this paper. However, worth mentioning is the recent paper of Leung [26], who used an Eulerian kinematics point of view. The flow map then satisfies a set of time dependent advection equations and requires the sophisticated WENO/TVD (Weighted Essentially Non-Oscillatory / Total Variation Diminishing) algorithm to achieve high order accuracy. The Discontinuous Galerkin finite element formulation could also be used to solve this set of partial differential equations, with the added benefice of error estimation and mesh adaptation [27].

To motivate the definition of FTLE, let us consider two neighboring particles, initially at $\mathbf{x}_0$ and $\mathbf{x}_0 + \boldsymbol{\epsilon}$, with $\|\boldsymbol{\epsilon}\|$ being a very small distance. As the two fluid particles follow their respective pathlines, they may tend to either stay very close to each other or to separate. The final separation $\boldsymbol{\delta}_{\mathbf{x}_0}(\boldsymbol{\epsilon})$ of the particles after a finite integration time T is approximated by a first order Taylor series expansion:

$$\boldsymbol{\delta}_{\mathbf{x}_0}(\boldsymbol{\epsilon}) = \boldsymbol{\varphi}_{t_0}^T(\mathbf{x}_0 + \boldsymbol{\epsilon}) - \boldsymbol{\varphi}_{t_0}^T(\mathbf{x}_0) = (\boldsymbol{\epsilon} \cdot \nabla)\boldsymbol{\varphi}_{t_0}^T(\mathbf{x}_0) + O\left(\|\boldsymbol{\epsilon}\|^2\right). \tag{5}$$

Thus, a simple linearized approximation of the fluid particles separation is given by $\|\boldsymbol{\delta}_{\mathbf{x}_0}(\boldsymbol{\epsilon})\| \approx (\boldsymbol{\epsilon}, \mathcal{S} \cdot \boldsymbol{\epsilon})^{1/2}$, with $\mathcal{S}$ being Cauchy-Green tensor (Eq. (3)). Hence, an upper bound to this estimate can be written as

$$\|\boldsymbol{\delta}_{\mathbf{x}_0}(\boldsymbol{\epsilon})\| \leqslant \|\boldsymbol{\epsilon}\| \|\mathcal{S}\|^{1/2} \equiv \|\boldsymbol{\epsilon}\| (\lambda_{\max}(\mathbf{x}_0))^{1/2}.$$

This ultimately results in the definition of the Finite Time Lyapunov Exponent (FTLE):

$$\sigma_{t_0}^T(\mathbf{x}_0) = \frac{\ln(\lambda_{\max}(\mathbf{x}_0))}{2|T|}. \tag{6}$$

This is a scale measure of the maximum linearized growth rate of the distance between initially adjacent fluid particles, tracked over a finite time interval. The FTLE is used to identify attracting or repelling material lines when the integration proceeds backwards or forwards in time respectively. Specifically, these lines of interest are *LCS*, and they correspond to local maximum of $\sigma_{t_0}^T(\mathbf{x}_0)$, or ridges, if we take the point of view of a 3D function graph. The *LCS* are then locally characterized by the eigenvalues of the Hessian of the FTLE field, whose existence are subjected to the following subsidiary hypotheses:

Hypothesis 2.4. *By observation, the Hessian of the FTLE field involves 3rd order derivatives of the flow map. This requires at least a $C^3(D)$ velocity field.* □

Hypothesis 2.5. *An eigenvalue field, such as $\lambda_{\max}(\mathbf{x}_0)$, is differentiable if it is simple. Since the FTLE field is a function of the largest eigenvalue of the Cauchy-Green tensor $\mathcal{S}$ (Eq. (3)), it is thus also assumed that $\lambda_{\max}(\mathbf{x}_0)$ is simple for all $\mathbf{x}_0 \in D$, or at least in a neighborhood around the LCS. In [4], this takes the form of its third hypothesis ($\ln \lambda_{\min}(\mathbf{x}_0) < 0 < \ln \lambda_{\max}(\mathbf{x}_0)$) when 2D flows are considered. This has been related to the notion of finite time hyperbolic trajectories, but not to the eigenvalue differentiability.* □

3. Computation of the spatial gradient and Hessian of the flow map

We target analysis and flow structure extraction from velocity field measurements that usually result in a very large database. Particle image velocimetry (PIV) is probably the most widely used technique [28] to obtain such data; however, to compute FTLE fields, measurements must be time-resolved, i.e. the acquisition frequency has to be sufficiently high to follow the flow evolution over time. In gas, only large-scale structures, such as coherent structures in turbulent flows, can be time-resolved, while in liquids, all velocity scales can sometimes be resolved.

Standard measurements generally provide two velocity components in a plane, and the fluid domain D is typically represented by a rectangular window. The velocity field is thus discrete and defined at the nodes of a uniform Cartesian mesh that covers the window. First and 2nd order partial derivatives of the velocity field are approximated by 2nd order finite difference approximations at the mesh nodes. This allows efficient bilinear interpolation of the velocity field and its derivatives within the mesh.

In Shadden et al. [4], the mesh nodes are assumed to be the starting coordinates of trajectories computed with (Eq. (1)), over a finite time T, to yield the flow map. Hence, the velocity field and the flow map are always known at the same nodes.

Again, 2nd order finite difference formulas are then appropriate to estimate the first derivatives of the flow map and the Cauchy-Green tensor $\mathcal{S}$. However, to capture the *LCS* within the FTLE field accurately, a large number of nodes is required. The mesh is then uniformly refined to capture the flow structures accurately, since they evolve and move continuously within the fluid domain D. For instance, in [29], at least 200,000 nodes (and sometimes up to 300,000 nodes) were necessary to capture a 2D Lyapunov field with accuracy.

When tackling large database, it is certainly inefficient to iteratively increase the number of nodes in a Cartesian mesh by blindly doubling the number of rows and columns (and planes in 3D). The velocity field has to be interpolated into this denser mesh without any increase in its accuracy, since we are working on measured data. It is thus much more efficient to work with two meshes: a structured uniform Cartesian mesh to support the discrete velocity field, and an unstructured non uniform mesh to optimally distribute the Lagrangian particles near flow structures that exhibit large gradients. However, in adopting this strategy, the nodal computation of the gradient and the Hessian of the flow map rely on finite element reconstruction techniques [11,30,31] that are not as cost effective or as accurate as finite difference formulas on a regular mesh. An alternative is to compute its gradient and the Hessian (when needed) concurrently with Eq. (1) by taking the 1st and 2nd order partial derivatives of this set of equations. This yields the following set of ordinary differential equations (Eqs. (7) and (9)) with their appropriate initial value. In the derivation of these equations, it is assumed that space and time derivatives commute.

Definition 3.1. The ordinary differential equation of the directional derivative of the flow map that satisfies the hypotheses of Definition 2.1 is defined as:

$$\frac{d}{dt}(\mathbf{v}\cdot\nabla)\boldsymbol{\varphi} = (\mathbf{v}\cdot\nabla)\mathbf{u}(\boldsymbol{\varphi},t) \tag{7}$$

along any unit and constant vector $\mathbf{v}$. For completeness, we also write this in Cartesian coordinates:

$$\begin{aligned} &\frac{d}{dt}\left(\frac{\partial\varphi_i}{\partial x_j}\right) = \sum_k \frac{\partial u_i}{\partial x_k}\Big|_{(\boldsymbol{\varphi},t)} \frac{\partial\varphi_k}{\partial x_j}\Big|_t \\ &\frac{\partial\varphi_i}{\partial x_j}\Big|_{t_0} = \delta^i_j, \quad \forall \mathbf{x}_0 \in D. \end{aligned} \tag{8}$$

As an example, the initial conditions reflect the fact that the directional derivative of the x coordinate in the x direction is 1, but zero in the y and z directions. The components of $(\mathbf{v}\cdot\nabla)\boldsymbol{\varphi}$ are $C^1(D)$. □

Definition 3.2. The ordinary differential equation of the Hessian of the flow map that satisfies the hypotheses of Definition 2.1 is defined as:

$$\frac{d}{dt}\mathcal{H}(\boldsymbol{\varphi}) = \mathcal{H}(\mathbf{u}(\boldsymbol{\varphi},t)) \tag{9}$$

and we write this in Cartesian coordinates for completeness:

$$\begin{aligned} &\frac{d}{dt}\left(\frac{\partial^2\varphi_i}{\partial x_j\partial x_k}\right) = \sum_r \left(\frac{\partial u_i}{\partial x_r}\Big|_{(\boldsymbol{\varphi},t)} \frac{\partial^2\varphi_r}{\partial x_j\partial x_k}\Big|_t\right) + \sum_r\left(\sum_s\left(\frac{\partial^2 u_i}{\partial x_r\partial x_s}\Big|_{(\boldsymbol{\varphi},t)}\frac{\partial\varphi_s}{\partial x_k}\Big|_t\right)\frac{\partial\varphi_r}{\partial x_j}\Big|_t\right) \\ &\left.\frac{\partial^2\varphi_i}{\partial x_j\partial x_k}\right|_{t_0} = 0, \quad \forall \mathbf{x}_0 \in D. \end{aligned} \tag{10}$$

Note that, in Cartesian coordinates, the initial conditions reflect the fact that 2nd order partial derivatives of any coordinate is zero. The components of $\mathcal{H}(\boldsymbol{\varphi})$ are $C^0(D)$. □

Using Eqs. (1), (7) and (9) concurrently has obvious advantages and disadvantages, with regard to the accuracy of the computation, the computational cost, and the complexity of the tools that have to be integrated. The direct benefit is that the flow map, its gradient, and Hessian can then be interpolated continuously on the unstructured mesh and approximated by P^k Lagrange polynomials [32] at the element level, depending on the number of nodes that define the element vertices. In this paper, we have chosen linear Lagrange polynomials for simplicity, and also because anisotropic mesh adaptation on this type of elemental approximation has reached maturity. However, the downside is the increase in numerical complexity and computational cost, since we must solve a set of 12 ordinary differential equations in 2D, or 30 in 3D. To be competitive, it has to be used in a mesh adaptation framework that provides the means to detect the large gradients within the flow map, to evaluate the discretization error, and to optimize the mesh in order to minimize this error. This is the topic of the next section.

4. Anisotropic grid adaptation

Mesh adaptive methods [11] assess the quality of an initial solution obtained on a coarse mesh by using some form of error estimation, and modify the structure of the numerical approximation in a systematic fashion to improve the overall

quality of the solution. Adaptive remeshing was selected because it provides the greatest control of element size and grading [24] to accurately resolve flow features [33–36] such as shear layers, stagnation points, jets, and wakes. In this approach, the problem is first solved on a coarse grid to roughly capture the physics of the flow. The resulting solution is then analyzed to determine where more grid points are needed, and an improved mesh is generated. The problem is solved again on the new mesh. This process is repeated until the required level of accuracy is achieved.

In this framework, the numerical solution is the flow map computed with Eqs. (1), (7) and (9), and the numerical approximation is related to the time stepper accuracy, the construction of the first and second derivatives of the flow field, the interpolation of the flow field and its derivatives, and, obviously, the number and density of Lagrangian particles (or nodes). Clearly, where we expect the gradient of the flow map to be high, the particle density should be increased, and, conversely, it should be reduced where the gradient is small, in order to reduce cost. Locally the directional derivative of the flow map may be maximal in a preferred direction, thus requiring non homogeneous particle distribution. The mesh connecting these particles is anisotropic, and can be optimized to yield the minimum interpolation error.

The construction of an anisotropic mesh is usually achieved by modifying how the distance between two neighboring points is measured [37]. This is realized by the construction of a metric field, such that the distance between the vertices of a simplex (triangle, tetrahedron) is unity. This technique belongs to the family of continuous metrics. It was introduced by Frey and George [37] and further developed by Alauzet [12] and Loseille [23]. Specifically, the methodology elaborated by Alauzet and Loseille defines an optimal metric that yields a finite element mesh, with a target mesh complexity, minimizing the interpolation error. The concept of mesh complexity is related to the number of nodes or, in our context, the number of Lagrangian particles. The mesh is not unique and depends on the L^p norm used to measure the interpolation error [14]. This technique is interesting, since the choice of the appropriate L^p norm defines the scale of the feature to be captured by the mesh. Typically, the L^1 norm provides meshes best suited to capturing small scale structures, while the L^∞ norm provides meshes optimized to capture large scale structures [12]. This means that the mesh density near large scale structures will be greater with the L^∞ norm than with the L^1 norm for the same mesh complexity, which is why we chose this technique for this paper. We use it to optimize the mesh in the vicinity of *LCS*.

To understand the methodology of Alauzet and Loseille, we begin by considering a scalar function $f : \Omega \subset \mathbb{R}^n \mapsto \mathbb{R}$, at least of class C^2. A continuous linear finite element of this function is then constructed by tiling Ω with a mesh of triangle (or tetrahedral) to minimize the interpolation error in the L^p norm. Obviously, a constraint has to be placed on the number of nodes, or equivalently the mesh complexity, since the solution of this problem would be for an infinite number of nodes. For this problem, the optimal metric that yields the optimal mesh is defined as follows:

Definition 4.1. The construction of the optimal continuous metric is based on the assumption that the function f, approximated by a piecewise linear approximation, is of class C^2. This yield the metric $\mathcal{M}_f^{\text{opt}}$.

$$\mathcal{M}_f^{\text{opt}}(\mathbf{x}) = D_{L^p} \det(|\mathcal{H}(f)|)^{\frac{-1}{2p+n}} |\mathcal{H}(f)|, \tag{11}$$

$$D_{L^p} = \frac{N^{2/n}}{\left(\int_\Omega \det(|\mathcal{H}(f)|)^{\frac{p}{2p+n}} d\Omega\right)^{2/n}}, \tag{12}$$

$$|\mathcal{H}(f)| = R^t \begin{pmatrix} \max\{|\lambda_1|, \epsilon\} & 0 & \cdots \\ 0 & \max\{|\lambda_2|, \epsilon\} & \cdots \\ \vdots & 0 & \ddots \end{pmatrix} R \tag{13}$$

with λ_i being the eigenvalues of the $\mathcal{H}(f)$ bounded by a small number ϵ, in order to guarantee that the construction of the metric is definite positive. In these relationships, N is the mesh complexity, p the parameter of the L^p norm, and n the dimension of Ω.

For completeness, we note that the mesh complexity can be directly computed from any metric field $\mathcal{M}$ from

$$N = \int_\Omega \sqrt{\det(\mathcal{M})} d\Omega, \tag{14}$$

where $\sqrt{\det(\mathcal{M})}$ is nothing less than the mesh density, or, in our context, a measure of the Lagrangian particle density. It was observed by Loseille that the mesh complexity is, in fact, proportional to the number of nodes in the mesh for a specific problem. □

Classical interpolation theory [38] usually relates the interpolation error to the maximum element size h. For a linear interpolation, we find that

$$\|f - \pi f\|_{L^p(\Omega)} \leqslant C h^2 \tag{15}$$

with C a constant that is independent of the solution and h. This simply implies that, within the asymptotic range, the global error is divided by 4, if all edges are uniformly divided by 2. In a 2D triangular mesh, the elements are divided into four congruent triangles. However, for tetrahedral meshes, this is not as easily accomplished, and, in general, when an unstructured

mesh is used, the maximal size of the elements is not an adequate parameter for monitoring solution quality. The parameter of choice is then the mesh complexity N, and the metric (Eq. (11)) is used to build a sequence of meshes with the same orientation and anisotropic coefficients that satisfy the following error estimate [23]:

$$\|f - \pi f\|_{L^p(\Omega)} \leqslant \frac{C}{N^{2/n}} \tag{16}$$

with C being independent of the solution and N. It is interesting to note that Akin [11] suggested that $h \approx 1/N_v^{1/n}$, with N_v being the number of nodes. This would certainly yield a similar result to Eq. (16) since $N_v \propto N$. However, Eq. (16) is valid only if the meshes are built with an optimal metric, a fact unspecified by Akin. For instance, if the partial derivatives of f are constant in the y direction, increasing the grid density in that direction will not translate in any global error reduction. This situation is avoided if an optimal metric is used.

The generalization of the metric driven mesh adaptation paradigm to vector value function $\mathbf{f} : \Omega \subset \mathbb{R}^n \mapsto \mathbb{R}^m$, is straightforward. We observe that to each individual components f_i we can define a continuous optimal $\mathcal{M}_{f_i}^{\text{opt}}$ with the above-mentioned methodology (Definition 4.1). However, we must combine all these metrics into a single one in order to build a single mesh. This is achieved through metric intersection [37], which is symbolically written as $\mathcal{M} = \mathcal{M}_{f_1}^{\text{opt}} \cap \mathcal{M}_{f_2}^{\text{opt}} \cap \cdots$. From a geometric point of view, metric intersection consists of the construction of the biggest ellipse that can be included at the intersection of two ellipses. If we have more than two metrics to intersect, the solution is not unique. In our 3D applications, we have elected to intersect the metric of the first component with the metric of the second component. The result was then intersected with the metric of the third component. When three metrics intersect, the mesh density and local element stretching depend on the order of the intersection. However, we have not encountered any loss of accuracy or reduction in the convergence rate when one order of metric intersection is favored over another.

We recall from classical linear interpolation theory [38] that the interpolation error at any points within an element K is written as

$$(f_i - \pi f_i)(\mathbf{x}) = -\sum_{j=1}^{n_v} L_j(\mathbf{x}) R_j(\mathbf{x}), \tag{17}$$

$$R_j(\mathbf{x}) = \frac{1}{2}\left(\mathbf{a}^j - \mathbf{x}\right)^t \mathcal{H}(f_i)(\boldsymbol{\zeta}(\mathbf{x}))\left(\mathbf{a}^j - \mathbf{x}\right) \tag{18}$$

with n_v the number of Lagrange linear interpolation functions L_j, and $\mathbf{a}^j$ being the element vertices, and $\boldsymbol{\zeta}(\mathbf{x}) \in K$. We note that the metric-based adaptation relies on some approximation of the Hessian known only at the mesh vertices. Then, the following approximation is used:

$$\mathcal{H}(f_i)(\boldsymbol{\zeta}(\mathbf{x})) \approx \sum_{j=1}^{n_v} L_j(\mathbf{x}) \mathcal{H}(f_i)_j \tag{19}$$

with $\mathcal{H}(f_k)_j$ being the vertices value of the Hessian.This allows a simple estimation of the interpolation error $\|\mathbf{f} - \pi\mathbf{f}\|_{L^2(\Omega)}$ (see Remark 4.2) that will be used to report grid convergence as a function of the number of nodes in our numerical experiments.

Remark 4.1. In practice, L^∞ metrics are approximated by setting p in (Eq. (11)) to a sufficiently large number. In our numerical experiments, we have selected $p = 100$. However, in [39], we used a value as low as $p = 10$ without any significant modification to the mesh density or to the local element stretching.

Remark 4.2. In theory, we should compute and report error estimation with the same norm used by the error model that lead to the construction of the L^p metric. However, since the flow map is $C^2(D)$, all L^p norms with $(1 \leqslant p < \infty)$ are equivalents and we can select the most convenient one. For our purpose, we have selected the L^2 norm since its integrand is smooth and requires the least number of Gaussian points for accurate integration.

Remark 4.3. We anticipate, from Eq. (16), a quadratic convergence rate when the error estimation is properly written as a function of $1/N^{1/n}$. However, it is usual to study grid convergence as a function of N the mesh complexity (or the number of nodes). Accordingly, the slope of the regression curve would then read $(p = -1)$ or $(p = -2/3)$, respectively in 2D and 3D, rather than $(p = 2)$.

4.1. Adaptation on the flow map

To improve the definition of ridges (or *LCS*) of the FTLE field through mesh adaptation, the most natural choice is to select $f(\mathbf{x}) = \sigma_{t_0}^T(\mathbf{x})$ and to build an optimal metric $\mathcal{M}_\sigma^{\text{opt}}$. This strategy has been studied by Bélanger [39] and was found to be optimal for capturing ridges of the FTLE field. In fact, we recall from Shadden [4] that one eigenvector of $\mathcal{H}(\sigma_{t_0}^T)$ is normal to the ridge and the other (in 2D) tangential to the ridge. Across the ridges, the gradient of FTLE field is extremely steep. The opti-

mal metric would then yield an anisotropic mesh with the longest edge almost tangential to the ridge. However, this approach requires computation of the second derivatives (Hessian) of the FTLE field at the nodes on an unstructured mesh. Well-known recovery techniques, such as the Zhu-Zienkiewicz [8,9], can be used for this purpose, but with an increase in computational cost. Furthermore, it also assumes that the velocity field is at least C^3, and that the maximum eigenvalue of the Cauchy-Green tensor is at least of class C^2.

In this paper, we study an alternative metric based on the flow map, such that $\mathbf{f}(\mathbf{x}) = \boldsymbol{\varphi}(\mathbf{x})$. This choice was inferred from the empirical observation that *LCS* are localized near local maxima of $\|\nabla\boldsymbol{\varphi}(\mathbf{x})\|$. In 3D, this approach results in three independent metrics that are intersected through application of the procedure described previously. The Hessian of the flow map is directly computed from Eq. (9) and does not use any recovery technique based on unstructured mesh. The velocities are known at the nodes of a simple structured mesh, which simplifies the construction of their derivatives. Our numerical experiments, both analytical and based on flow field measurements, show that this metric is capable of detecting *LCS* of the FTLE field.

5. Two-dimensional analytical examples

The efficiency of the procedure described in the previous sections is studied with analytical flow field of class C^∞ in space and in time. The velocity field is defined in $U \equiv \mathbb{R}^n$, and the particles are allowed to move freely in U when the mesh is adapted to the time-evolving structures in the computational domain $D \subset U$. Consequently, the flow field satisfies hypotheses (2.1-2.3) and yields a well defined FTLE field. In the numerical experiments, the velocity field and its derivatives are computed from analytical functions to avoid the introduction of any bias introduced by interpolation and reconstruction of derivatives from a discrete approximation of the velocity field. This implies that the pointwise solutions of Eqs. (1), (7) and (9) depend only on the accuracy of the time-stepper. To reduce this error to a minimum, we have selected Butcher's 5th order explicit Runge–Kutta algorithm [40].

Since we are using a finite number of particles to define a mesh, the global error is a function of $E \approx O(\Delta t^5, 1/N^{2/n})$. The first term corresponds to the integration error over time, while the second results from the linear finite element interpolation of the flow map. If the time step is well chosen, the discretization error over time becomes negligible with regard to the finite element approximation, and we should expect $E \propto 1/N^{2/n}$, at each time step, when computing the interpolation error with the procedure described in Section 4. This implies a quadratic convergence rate of the interpolation error.

Under these optimal conditions, we will specifically examine, in this section, the reliability of the mesh adaptation software of adapting to the FTLE field through our proposed error estimate of the flow map. Two analytical examples will be used for this purpose in Section 5.1 and 5.2

5.1. Solomon and Gollub's 2D analytical Rayleigh-Bénard model

Solomon and Gollub [41] describe a simplified 2D model of time-periodic Rayleigh-Bénard convection transport, where the velocity field is derived from a stream function

$$\begin{aligned} &\psi(x,y,t) = \sin\left(\pi(x - g(t))\right)\sin(\pi y),\\ &g(t) = 0.3\sin(4t) + 0.1\sin(2t), \end{aligned} \tag{20}$$

with $g(t)$ an oscillatory function that generates moving *LCS*, the length and width of which vary according to the integration time T. The specific parameters used in Eq. (20) are such that the flow is periodic in x

$$\mathbf{u}(x,y,t) = \mathbf{u}(x + 2n, y, t) \quad \text{with} \quad n \in \mathbb{Z} \tag{21}$$

and is bounded by two time-invariant streamlines, at $y = 0$ and $y = 1$, which restrict the motion of the *LCS*. Accordingly, the computational domain D is defined in the following manner to capture all flow structures:

$$D = \{(x,y) \in \mathbb{R}^2 \,|\, (0 \leqslant y \leqslant 1)\ \&\ (0 \leqslant x \leqslant 2)\}. \tag{22}$$

In our numerical experiments, the starting time ($t_0 = 0.5$ sec) and the integration time ($T = 1$ sec) were chosen in order to enable comparison of our results with the isotropic refinement method of Lekien and Ross [7]. The results, summarized in Fig. 1, were computed with $\Delta t = 1/75$ sec.

As a gold standard, we have computed the FTLE field on a regular Cartesian grid with 20,000 nodes (200 by 100 nodes) with the original methodology of Shadden et al. [4]. Fig. 1(a) is a 2D view of the FTLE field, while Fig. 1(b) is a 3D projection of it. In the latter representation, the *LCS* clearly stands out as ridges of this field, and we see that the finite difference approximation of the flow map derivatives on this grid smooth out the FTLE field, unlike the grid converged solution of Fig. 1(e). In contrast, the FTLE field in Fig. 1(c) was computed on the same grid with Eqs. (1), (7) and (9), and constitutes our proposed alternative, as it solves the derivatives along the particles pathlines. We observe immediately that the uniform grid distribution on the boundaries is fine enough to accurately capture the intensity of the FTLE field. Moreover, we observe that the point wise values of the FTLE field are also well computed within the computational domain, as anticipated by our theory. The benefit of a metric-based L^∞ anisotropic grid adaptation is revealed in Fig. 1(d), which shows a grid of 20,394 optimally distributed nodes. This grid, with almost the same number of nodes as our gold standard, is able to capture the

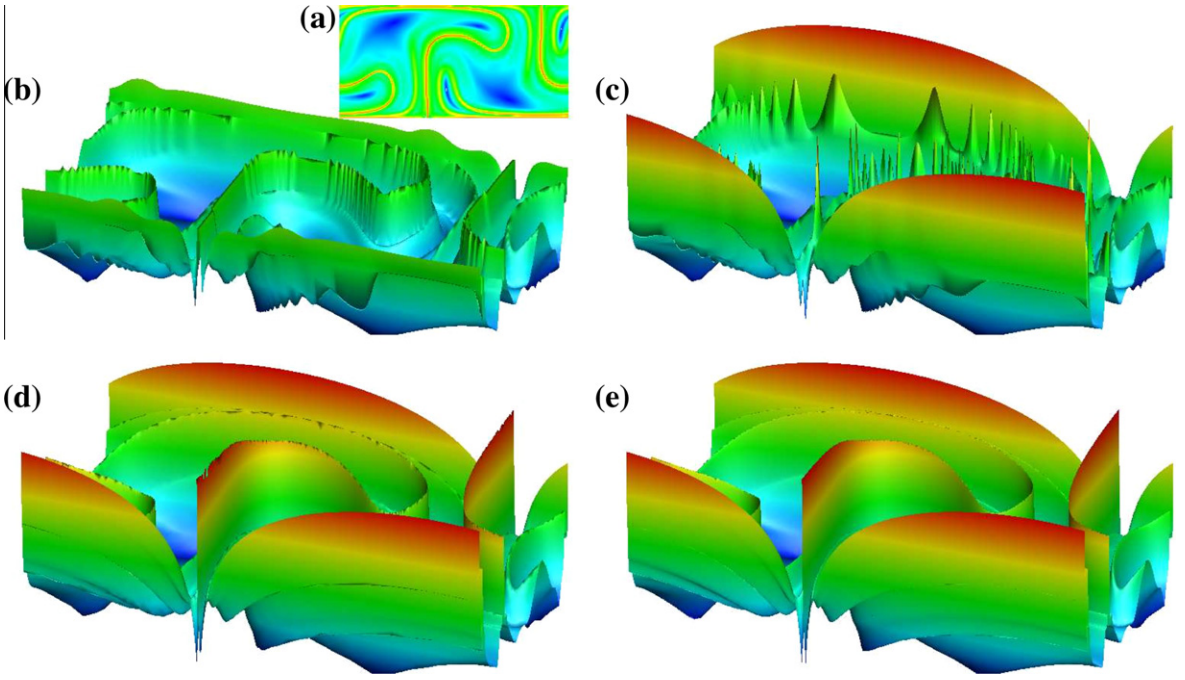


Fig. 1. Solomon and Gollub's flow field: (a) and (b) FTLE field computed with the methodology of Shadden [4] on a 20,000 nodes Cartesian mesh, (c) FTLE field computed with (Eqs. (1) and (7) on a 20,000 nodes Cartesian mesh, d) L^∞ metric-based adaptation with 20,394 nodes, (e) L^∞ metric-based adaptation with 104,687 nodes.

intensity of the FTLE field accurately throughout the computational domain. However, a close look at the ridges reveals small sawtooth patterns that are a consequence of the large gradients of the flow map in the neighbourhood of the *LCS*. This is corrected by choosing a smaller integration time T, which would smooth out the ridges, or by increasing the complexity (or equivalently the number of nodes) of the anisotropic adapted mesh, as shown in Fig. 1(e), where there are 104,687 nodes.

In Fig. 1(d), the mesh has 38,236 anisotropic elements. This is approximatively 30% fewer than the best isotropic adapted mesh of Lekien and Ross [7] for a comparable FTLE field representation. It is important to stress that they have not provided any error estimation method related to their adaptation process. Hence, without the benefit of a common error estimation method, it is difficult to quantitatively compare these two adaptation methodologies as applied to FTLE field reconstruction. However, the accumulated body of knowledge in grid adaptation [37] strongly indicates that, with equal mesh complexity, the anisotropic mesh error is less than the isotropic mesh error and so yields a better solution.

In Fig. 2, we show the anisotropic mesh, coloured according to the FTLE field, adapted to the flow map of Fig. 1(d). We see that the elements are denser across the ridges and elongated in the tangential direction, as we have anticipated. So, the adaptation on the flow map is empirically equivalent to the adaptation on the FTLE field. In the lower-left caption, the details of the mesh in this zone are shown, and the mesh is further enlarged in the lower-right caption. These captions show that the elements are highly stretched in the vicinity of the flow structures, with a typical anisotropic ratio $r = h_x/h_y = 1/350{,}000$. Increasing the integration time T requires smaller elements across the ridges with a higher aspect ratio. We must point out that we have sometimes experienced an anisotropic ratio $r = h_x/h_y = 1/1{,}000{,}000$, which reduces the reliability of our metric-based adaptation software [25].

In Fig. 3, the mesh topology is shown as a function of the number of nodes, and also in relation to the L^p norm used in the metric construction. In the first four upper captions, the L^∞ metric was used to detect large *LCS* structures. In contrast, the lower-left caption shows that the L^1 metric is suited to the detection of small scale structures and distributes nodes in regions bounded by large structures. Finally, the last lower-right caption is simply a log-log graph of the interpolation error of the flow map versus N. The slope of the regression curve is -1, and corresponds to a quadratic convergence, as predicted by the theory (see Remark 4.3). This also shows, indirectly, that the time step used was small enough to render the discretization error over time negligible.

5.2. Kelvin cat's eyes model

Kelvin cat's eyes flow [42] models the periodic row of vortices from the exact solution of the two-dimensional Euler equations. These vortices, centred at $y = 0$, develop within the layer bounded by two opposite parallel flows of constant magnitude. In this simplified model, the velocity field is derived from a stream function given by:

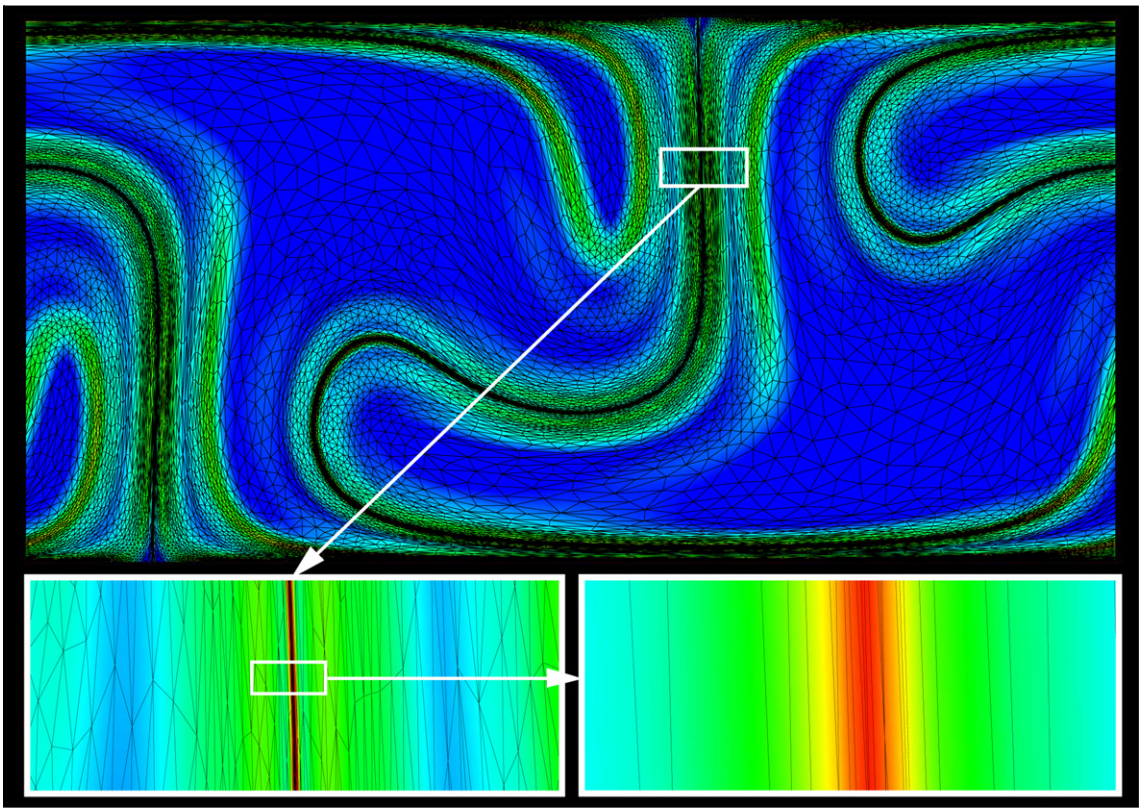


Fig. 2. Mesh adapted to Solomon and Gollub's FTLE field with a maximum anisotropic ratio of 1:350,000.

$$\psi(x,y,t) = -u_c\left[y + \frac{\lambda}{2\pi}\ln\left\{\cosh\left(\frac{2\pi}{\lambda}y\right) - a\cos\left(\frac{2\pi}{\lambda}(x-u_c t)\right)\right\}\right] \tag{23}$$

where a is a parameter that characterizes the vorticity concentration, u_c is the convection velocity of vortices, and λ is the distance between consecutive vortices. For $a = 1$, the solution reduces to point row vortices and for $a \ll 1$, to periodically perturbed parallel shear layer with a hyperbolic tangent profile. The specific parameters used in Eq. (23) are such that the flow is periodic in x

$$\mathbf{u}(x,y,t) = \mathbf{u}(x+n\lambda,y,t) \quad \text{with} \quad n \in \mathbb{Z} \tag{24}$$

with

$$\lim_{y\to+\infty}\mathbf{u}(x,y,t) = (2u_c,0) \quad \text{and} \quad \lim_{y\to-\infty}\mathbf{u}(x,y,t) = (0,0), \tag{25}$$

the maximum value of the velocity far away from the vortices' row. Accordingly, a suitable computational domain D is defined in the following manner to capture, at all time, the *LCS* induced by the vortices motion:

$$D = \{(x,y) \in \mathbb{R}^2 \,|\, (-0.5 \leqslant y \leqslant 0.5) \;\&\; (0 \leqslant x \leqslant 1)\}. \tag{26}$$

In our numerical experiments, the starting time ($t_0 = 2$ sec) and the integration time ($T = -0.6$ sec) were chosen. The results, summarized in Fig. 4, were computed with $\Delta t = 0.01$ sec, $a = 1/2,\ u_c = 0.5$, and $\lambda = 0.2$.

In Fig. 4(a) and 4(b) the mesh has 9,465 anisotropic adapted nodes to the FTLE field with a L^∞ metric-based adaptation procedure. As shown with the previous example, the anisotropic mesh is adapted to follow the ridges of the FTLE field. The longest side of the element are almost tangent to the ridge and the shortest one normal to the ridge to adapt to the local variation in the gradient of the FTLE field. This confirmed, empirically, the efficiency of the adaptation of Hessian-based flow map as a surrogate to the direct adaptation on the FTLE field itself. Furthermore, in Fig. 4(a), we have superimposed the relative velocity field to illustrate the property of *LCS* of being material lines with almost zero mass fluxes.We observe that the relative velocity vectors are tangent to the *LCS* (in red), and this implies almost zero mass fluxes through these structures [4]. A mesh of equivalent complexity is given in Fig. 4(c) to illustrate the relative redistribution of nodes, from the neighbourhood of the *LCS*, into the computational domain when the adaptation is driven by a L^1 metric. This reduces the accuracy of the *LCS*

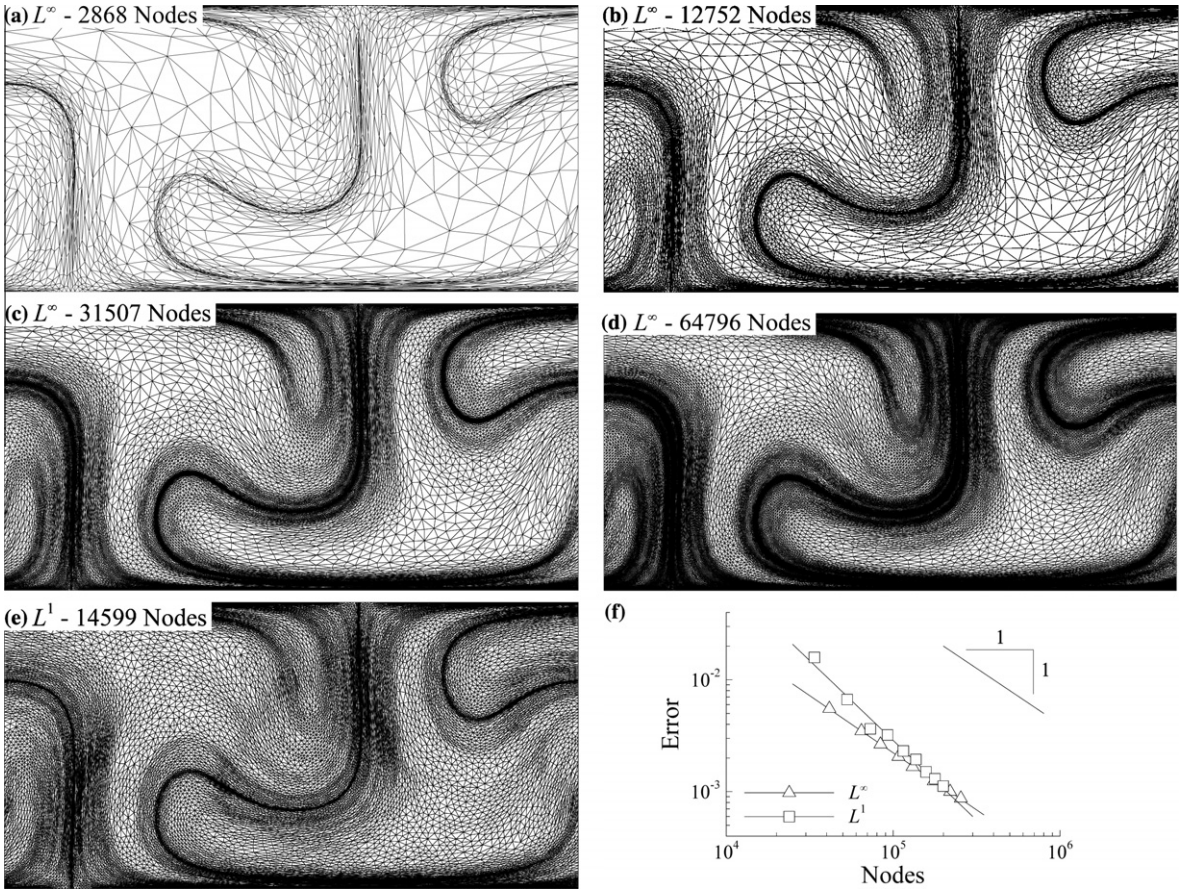


Fig. 3. Grid convergence analysis of Solomon and Gollub's flow map.

representation to the benefit of smaller structures. Finally, the last lower right caption, is a log-log graph of the interpolation error of the flow map versus N. As indicated, the two regression curves correspond to the grid convergence analysis with the L^1 and the L^∞ metrics. Their slopes correspond to a quadratic convergence in space (see Remark 4.3) with negligible discretization error in time.

6. Two-dimensional experimental examples

In this section, we apply the methodology described above to two-dimensional experimental data. Although the numerical procedure is similar, the measurement data present characteristics that are not considered by analytical examples. For instance, analytical solutions cannot represent complex unsteady flows, such as vortex dynamics and turbulence, where the temporal and spatial frequency content ranges from large- to very small-scale velocity fluctuations. Moreover, in practical situations, the velocity field is limited to the computational domain D, and the particle trajectories stop abruptly when they reach its boundaries. It is known that this may create artifacts [43] and modify *LCS* shape and intensity. In this case, grid adaptation does not yield a reduction in interpolation error as we increase the number of nodes, and this may be used as an indicator to proceed with care with the analysis of the flow physics. We will explore in Sections 6.1 and 6.2 two practical examples where this situation arises.

6.1. The impinging jet

This example is based on time-resolved PIV measurements of a circular jet impinging on a flat plate. The flow is laminar (the Reynolds number, based on the jet diameter and the exit velocity, is $Re = 1260$), and the fluid is water so that both space and time scales are fully resolved. The velocity field is known only within a finite measurement window $U \subset \mathbb{R}^2$ and, accordingly, the computational domain D is defined in the following way

$$D \equiv U = \{(x,y) \in \mathbb{R}^2 \,|\, (0 \leqslant y \leqslant 17.2\text{mm}) \,\&\, (0 \leqslant x \leqslant 52\text{mm})\} \tag{27}$$

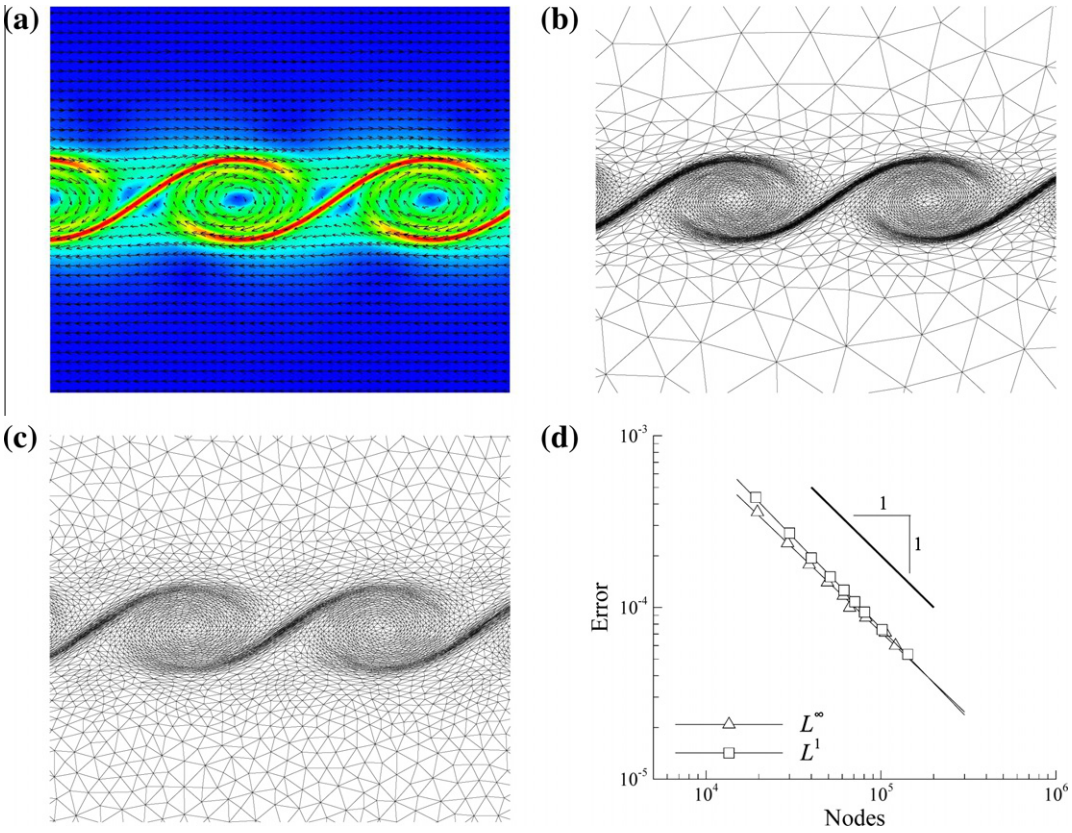


Fig. 4. Kelvin cat's eyes flow field: (a) adapted L^∞ metric FTLE field with 9,465 nodes and 19,075 elements, (b) adapted L^∞ metric mesh with 9,465 nodes and 19,075 elements, (c) adapted L^1 metric mesh with 9,771 nodes and 19,471 elements, (d) interpolation error estimation.

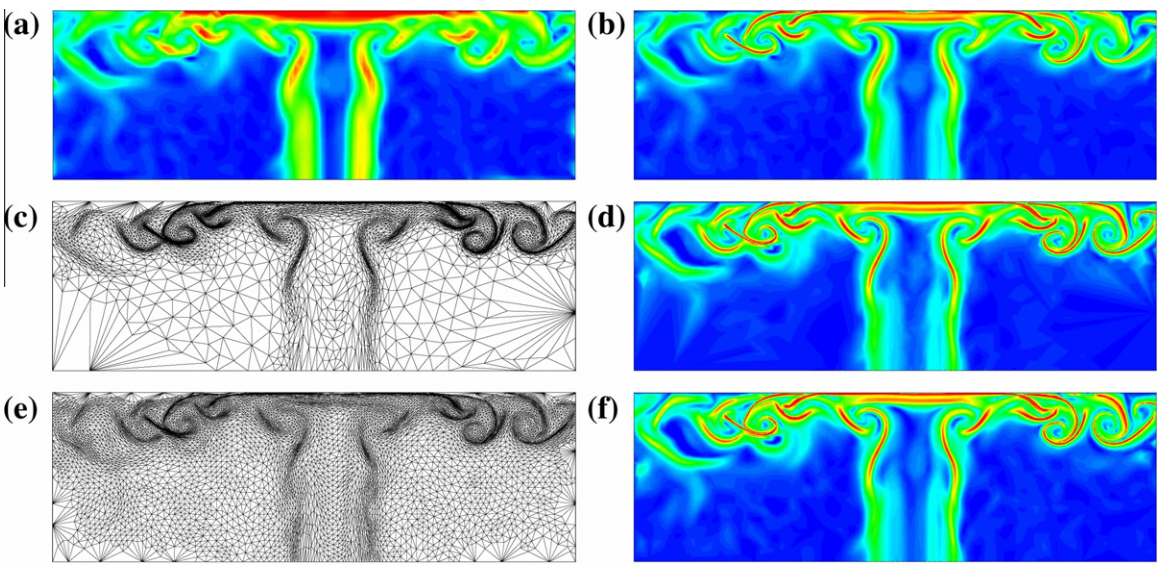


Fig. 5. FTLE fields computed from velocity measurements of a circular impinging jet (flow is from bottom to top). Results obtained for the initial PIV grid in (a) and the adapted mesh in (b) were computed with an integration time T that is 2/3 of that used to compute the results obtained with the L^∞ metric and the L^1 metric shown in (c), (d) and (e), (f), respectively.

to capture all flow structures within the measurement window. The results, summarized in Figs. 5 and 6, were computed with $\Delta t = 0.001$ s.

To detect the formation of vortical structures and to analyze their spatio-temporal evolution, FTLE fields are computed by integrating trajectories backward in time ($T < 0$ because fluid particles collect in forward times towards the vortex centers). The jet is centred and enters from the bottom boundary of the measurement window D, which coincides with the location of the nozzle lips, and then impinges on the wall positioned at around two jet diameters. In the lower-half of D, the boundaries of the jet are clearly identified by FTLE contours. The FTLE field, shown in Fig. 5(a), was obtained with the original PIV grid. The backward integration time ($T = -0.08$ s) was chosen to best reveal the flow structures, i.e. the birth of vortex rings inside the jet core and their development as they impinge on the wall. While it is possible to refine the FTLE fields by using longer integration times, this generally results in a degradation of the contours due to the coarse initial PIV mesh. In contrast, by using adaptive meshes, the computation shown in Fig. 5(b), obtained with the same integration time, reveals very fine flow details and, in particular, the formation of vortices on the wall resulting from the impingement of vortex rings appearing in the jet core. With this new grid, it is now possible to capture greater details by increasing the integration time ($T = -0.12$ s) as it is shown in Fig. 5(c) and (d), for the L^∞ metric, and in Fig. 5(e) and (f), for the L^1 metric, where structures are represented by clear and sharp patterns.

Higher values of the integration time result in longer particle trajectories and therefore a larger number of them leave the measurement domain. Most of these particles come, however, from the lower-half region of the measurement window. Since this region corresponds to the potential core of the jet, the measurement domain was extended by simply increasing the length of the potential core. For that, the inlet velocity profile, which is characterized by an almost constant velocity inside the jet and a zero velocity outside, was extrapolated in the upstream direction. The particles are then allowed to leave the computational domain and their position projected onto the bottom boundary of the measurement window to interpolate the velocity vectors and its derivatives. This results in a quadratic convergence in space (see Fig. 6(a)), as predicted by the theory (see Remark 4.3), and clearly implies that the structures identified within the measurement domain are meaningful and without undesirable artifacts. We have found that this was the simplest definition of the velocity field (and its derivatives outside the computational domain) that yield a reduction in the interpolation error when the grid was optimally refined. Finally, in Fig. 6(b), we show the grid convergence analysis of a subset of the measurement window located in the upper left quadrant that includes many near wall vortical structures. In this subset no particles left the measurement window, and the grid convergence analysis is once again quadratic.

The meshes shown in Fig. 5(c) and (e) are composed of approximatively 20,000 nodes, which represents 1/10th of the nodes that were required to reach the same level of details on a regular Cartesian grid. These results clearly demonstrate that the mesh adaptation increases the number of nodes close to the wall, where the vortex dynamics is complex, and on the contrary depletes it away where the flow is more quiescent. While the two metrics reveal the vortical structures apparently with the same details, the L^∞ metric concentrate fewer nodes in the upstream region of the flow compared with the L^1 metric. This leads, in the contour plot of Fig. 5(d), to a lower resolution of the FTLE field in the ambient fluid entrained by the jet in contrast to Fig. 5(f). While the analysis of unsteady flows is generally focused on the vortex dynamics revealed by *LCS*, some regions, although not always defined by ridges of the FTLE field, can help in the analysis of turbulent flow. This is examined in the next section.

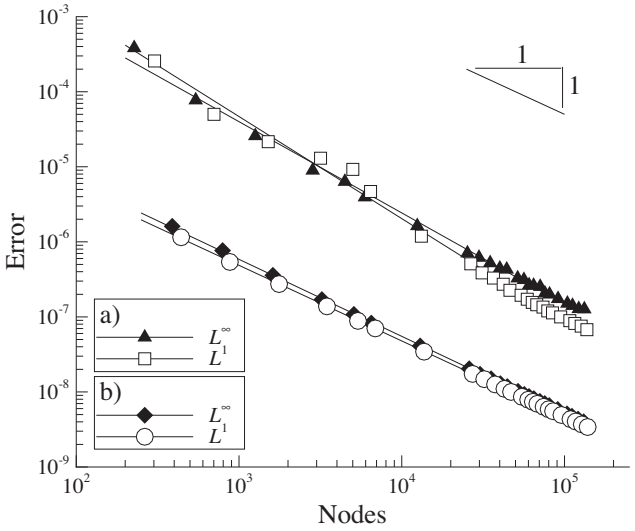


Fig. 6. Grid convergence analysis with L^1 and L^∞ metrics: (a) the computational domain covers completely the measurement window. The velocity field and its derivatives are extrapolated outside the computational domain to obtain an unbiased flow map; (b) the computational domain is a subset of the measurement window. The Lagrangian particles do not leave the measurement window and no extrapolation of the velocity field is necessary.

6.2. Transition to turbulence in a smooth axisymmetric constriction

The second experimental example concerns the flow generated from a local constriction in an axisymmetric conduit. The smooth decrease in the pipe diameter to half its size caused by the constriction is followed by an increase in this diameter back to its original size by a diffuser. The flow cannot remain attached, and separates in the diverging part of the constriction, which results in the development of an internal laminar jet surrounded by an axisymmetric recirculation region (the reader is referred to ([44]) for details). Fig. 7 shows the FTLE field computed from time resolved, two component PIV data obtained in a meridional plane immediately downstream of the constriction (Fig. 7(a) and (b)) and in a second plane juxtaposed to it (Fig. 7(c)). Each region covers a measurement window one diameter high and three diameters long.

The Reynolds number, based on the sectional average velocity in the unoccluded pipe and the diameter of the pipe, is $Re = 1160$. In Fig. 7(a), the presence of an axisymmetric shear layer is revealed, in the upstream region of the flow, by the FTLE contours that concentrate on the two horizontal straight lines marking the boundaries of the internal jet exiting the constriction, as was the case with the impinging jet (cf. Fig. 5). Here, the potential core is longer and extends to the right boundary of the measurement domain. Therefore, a large number of particles are lost during the computation of their trajectories, which explains the dark region in the middle of the FTLE field. However, the instability of the shear layer is evident in the birth and growth of vortex rings, and in the transition to turbulence occurring in the downstream part of the flow domain. Thus, the loss of particles does not, in general, present too severe a constraint, at least in vortical flows. Indeed, the integration time is often close to a time scale of the flow, such as the turn-over time of a vortex. If a particle leaves the domain, this is generally because it has not crossed a region of intense mixing and is therefore less important in the flow analysis. For instance, Fig. 7(b) shows an FTLE field obtained at the same location as in Fig. 7(a), but for another instant. The flow symmetry is broken, which triggers a more rapid transition to turbulence than previously. The first structure appearing in the lower part of the flow, at approximately one diameter downstream of the constriction, is clearly defined whereas the same region was not as sharp as in Fig. 7(a). In this last example, the asymmetric flow development results in the formation of bigger vortices, which decreases the length of the potential core and the size of the region where particles are lost during the computation of the FTLE. This region is particularly small in the FTLE field shown in Fig. 7(c), computed with velocity fields measured downstream of the transition. Results reflect the intense mixing generated by turbulence in this region. With more complex trajectories, the residence time of the particles inside the measurement domain is longer than in the examples in Fig. 7(a) and (b), and fewer particles are lost. However, while the procedure used in the previous section to increase the velocity data outside the domain can be used in the upstream direction where the flow is steady (Fig. 7(a) and (b)), it cannot be used in unsteady and turbulent flows (Fig. 7(c)).

One possible artifact in the FTLE field can originate from the time line that gathers at t_0 all particles that are on the domain boundaries at time $t_0 + T$. The nature of these artifacts and their impact on the flow map can be easily understood by considering a simple 1D flow of constant velocity $u > 0$ within a domain $D = [0, L]$. We arbitrarily split this domain into two by selecting $A \in D$ such that $T = (L - A)/u > 0$. Hence, if the particles are in $[0, A]$, then they freely flow along their paths, and, if they are in $]A, L]$, they stop abruptly at $x = L$ before completing their trajectories from time t_0 to time $t_0 + T$. Under these conditions, the solution to Eq. (1) is given by the following relationship:

$$\varphi = \begin{cases} x + L - A, & x \in [0, A] \\ L, & x \in]A, L] \end{cases} \tag{28}$$

By direct observation, we see that this is a class $C^0(D)$ solution, without the minimum global level of continuity required for the mesh adaptation and error estimation procedures to make sense. Nevertheless, the structures revealed through the adaptation process in $[0, A[$ will be physically meaningful, since the flow map exhibits the required local level of continuity. In the

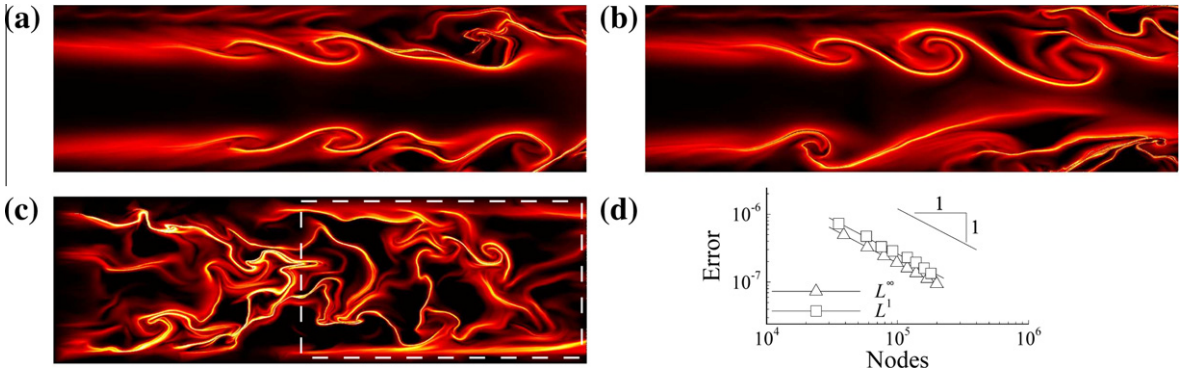


Fig. 7. FTLE fields computed in a meridional plane downstream of a smooth constriction in a pipe. (a) and (b) are located immediately downstream of the constriction while (c) is located three pipe diameters downstream (instantaneous velocity fields have not been acquired simultaneously and flow is from left to right). The error shown in (d) was computed with data located inside the domain indicated by the white box in (c).

other region, $]A, L]$, the structures will be artifacts of the numerical integration procedure. Finally, at $x = A$, the interface between these two regions, the FTLE field is discontinuous, in contradiction of the theory described in the previous sections.

To understand the artifacts generated by the limited measurement window in the FTLE field presented in Fig. 7(c), we performed a computation using only the data in the white box. We observed no significant artifacts, even though the intensity of the FTLE was lower in the left region of the domain than it was when we included all the data. For the turbulent flow considered here, the integration time is relatively short, compared with that of analytical solutions or laminar flows.

We tested a method similar to the velocity field continuation technique used in [43] to allow the particles to move freely outside the computational domain and to estimate their impact on the FTLE field. Although a slight improvement in some of the structures was observed, these changes were found to be negligible. So far, therefore, to our knowledge, there is no continuation technique that is proven, universal, and fail-safe which guarantees a $C^2(D)$ flow map within the computational domain D. Every technique has to be tailored, if possible, to the flow field to yield a grid independent FTLE field (i.e. interpolation error reduction). Note that the error shown in Fig. 7(d) was computed only from the data in the white box in Fig. 7(c).

Fig. 8 shows results obtained at the same location as in Fig. 7(c), but for another instant. This second example is presented because some regions of the FTLE field are less intense than in the first example, which is a consequence of the intermittency of the flow. Although transition to turbulence occurs behind the constriction, the low Reynolds number reached ($Re = 1160$) is less than the critical Reynolds number for transition in a pipe, so that the flow relaminarizes far downstream of the constriction. The right part of Fig. 8(a) shows the early beginning of this state. The instantaneous velocity profiles in this region can be decomposed into the flat, turbulent-like velocity distribution across the section of the pipe and a fluctuating component of small amplitude (the turbulence intensity is low in this region). Consequently, a pattern of fluid particles is convected without strong distortion, which results in coherent structures with a low and smooth imprint on the FTLE field, in contrast to the left part of Fig. 8(a), where the vortices are strongly deformed by an intense shear (the turbulence intensity is high). However, the FTLE field in the right region, which is not as marked as in the left region, provides remarkably similar features to those in visualizations obtained by injecting dye or smoke into experimental flows, and therefore should be adequately resolved by the FTLE computation for analysis purposes. While it could be possible to reveal more details by simply increasing T, the required value is so high that most of the particles leave the domain during the trajectory computation. Similar structures can be seen in [45], where the FTLE was computed from a direct numerical simulation database of a turbulent boundary layer, and also appear in the recirculation region that develops downstream of the constriction at the top of Fig. 7(a) (the visual imprint is very similar to the transport of a passive scalar).

The FTLE fields shown in Fig. 8 reveal that the level of detail that is reached depends on the norm chosen. While both norms are able to capture the *LCS* in the left part of the flow, it is clear that the node density obtained with the L^1 metric provides a better visualization of structures defined by the smallest FTLE values. This indicates that, in turbulent flows, the L^1 metric could reveal more than the L^∞ metric, which, in contrast, seems to be better adapted to laminar flows and to the extraction of separation/ attachment surfaces.

7. Three-dimensional examples

While the computation of 3D trajectories from experimental data has long been restricted by technological limitations of measurement techniques that mostly provide velocity components in planes, the current growth of tomographic PIV, providing the time-resolved, three components of velocity vectors in volumes, will allow the computation of 3D FTLE fields and thus will require an order of magnitude more nodes than in the 2D case and significant computing resources. As a consequence, mesh adaptation appears as a useful technique and in some cases even necessary from the point of view of massive data post-processing and visualization. This is illustrated with the following two examples.

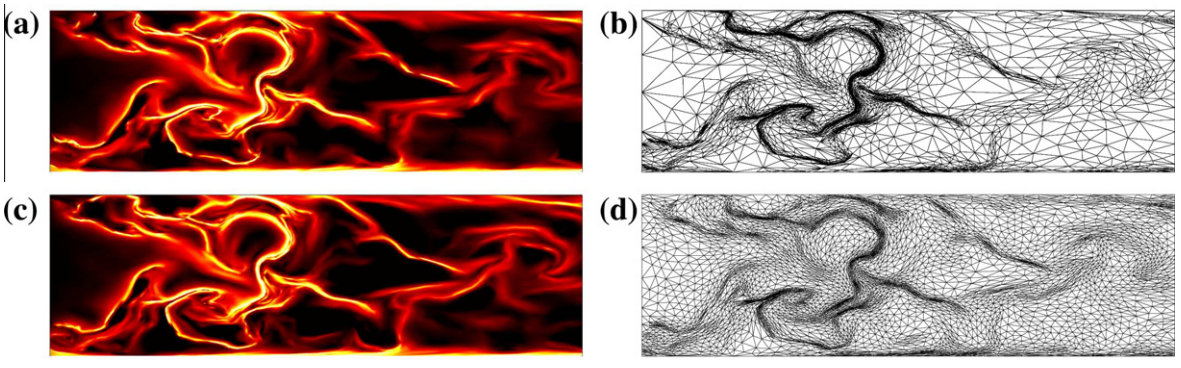


Fig. 8. FTLE fields computed 3 diameters from the constriction with their associated meshes. (a), (b) were obtained with the L^∞ metric and (c), (d) with the L^1 metric.

7.1. Three-dimensional extruded cat's eyes model

The Kelvin cat's eye model was also used to validate 3D metric adaptation computations. The 3D flow model was obtained by extruding the 2D mesh in the out-of-plane direction while preserving the velocity data and by setting the velocity in this direction to zero. In this extreme situation, the Hessian of each velocity component has at least one zero eigenvalue. The metric construction and intersection were then bounded by a small number, $\epsilon = 10^{-5}$, to render the metric definite positive (Eq. (13)). The 2D solution is compared to the 3D solution in Fig. 9(a) and (b) to illustrate that the quality of the FTLE field did not deteriorate, owing to the highly anisotropic mesh generated by the 3D L^∞ metric. In the small insert of Fig. 9(c), we observe that the mesh is highly stretched in the out-of-plane direction, as anticipated by the metric construction. Fig. 9(d) shows that the convergence rate of the interpolation error is close to a quadratic convergence, as theoretically predicted by Eq. (16).

7.2. Flow through a bileaflet mechanical heart valve

To the author's knowledge, the computation of FTLE fields has not been applied to experimental flows in 3D. To date, only 2D measurement windows have been considered, which provide the exact FTLE field in 2D flows or an approximation of it in 3D flows, since the out-of-plane movement of fluid particles is neglected in the latter. This is because of the difficulty of

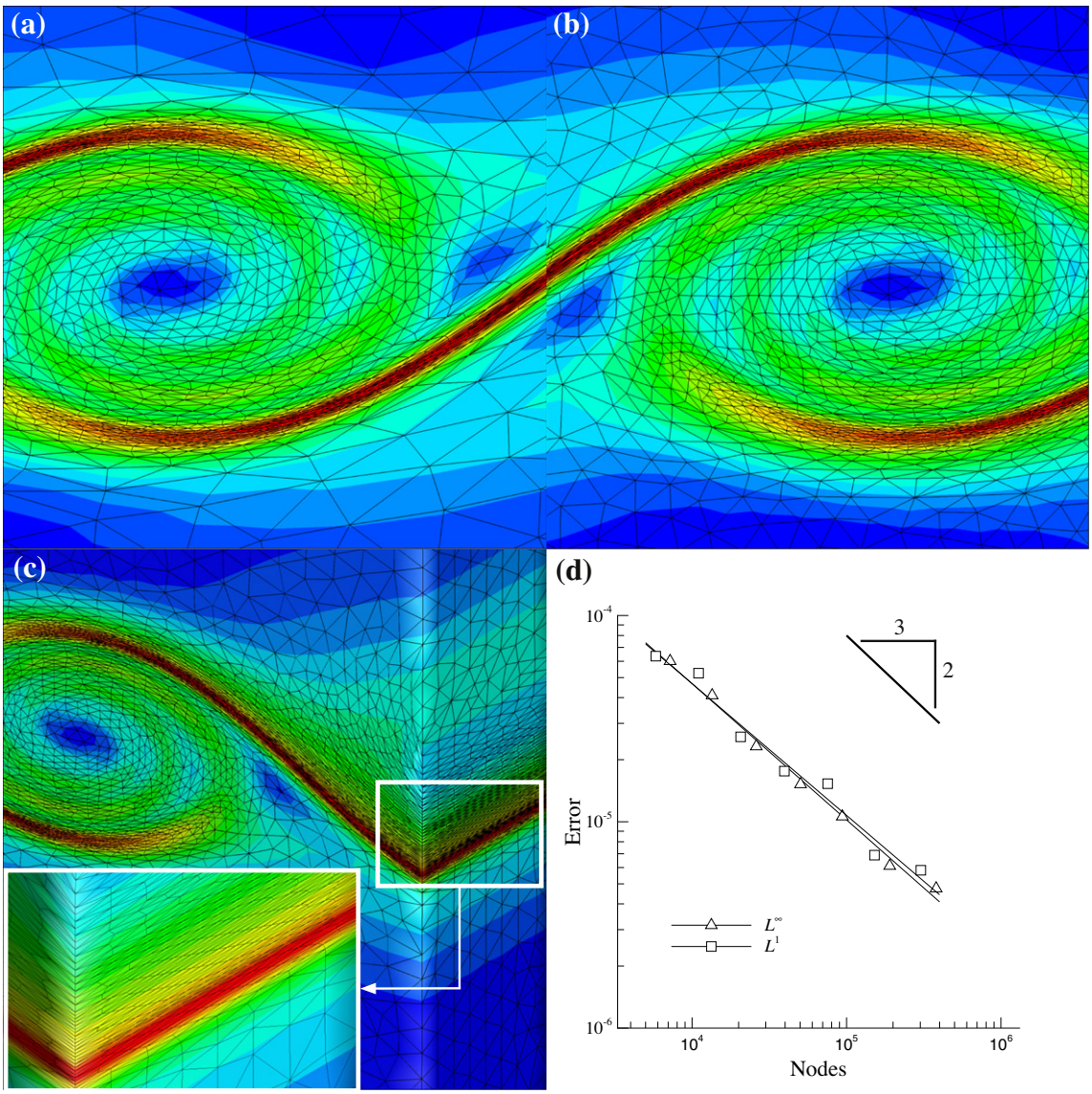


Fig. 9. Mesh adaptation of the Kelvin cat's eye flow in (a) 2D and (b), (c) 3D (L^∞ metric). The 3D case was obtained from an out-of-plane extrusion of the 2D velocity field, and the interpolation error estimation is shown in (d).

acquiring instantaneous 3D velocity fields in experiments, made more difficult by the requirement of a sufficiently high acquisition rate. These difficulties have been circumvented in this last experimental example by using a reconstruction technique to obtain 3D information. This was applied to the flow developing through a bileaflet mechanical heart valve installed in place of the smooth constriction previously studied (the geometry of the valve is shown in Fig. 10). For each flow pulsation, the two leaflets of the valve open and close, and, during the open phase, the inlet flow can be considered as quasi steady. The PIV stereoscopic technique [28] was used to measure the three components of the velocity vectors in a cross-sectional plane located at six pipe diameters from the valve. The time history of the flow was used to estimate velocity data in the out-of-plane direction by invoking the Taylor hypothesis. By assuming that the flow is frozen, the velocity field captured over time in a fixed reference plane can be interpreted as that induced by the advection of a 3D field traveling with the convection velocity u_c. Thus, the three components of the velocity vectors in a volume are known, as are their spatial derivatives (in particular, the spatial derivative in the out-of-plane direction z is estimated from the time derivative with $\partial_z = -u_c^{-1}\partial_t$). Finally, the 3D FTLE field can be computed from the resulting data volume by simply advecting the frozen velocity field in time in the streamwise direction using the equation $\mathbf{u}(z,t) = \mathbf{u}(z - u_c t, 0)$.

Our results are presented in Fig. 10. The measurement domain generated by the use of the Taylor hypothesis is a circular cylinder centered on the pipe axis and bounded by the pipe wall, so that analysis of the flow can only be obtained from 2D cuts of the 3D FTLE field. The cut through the symmetry plane perpendicular to the leaflets is shown in Fig. 10(a). The FTLE field detects the wake developing downstream of each leaflet, and, in particular, the top-wake strongly evokes visualizations of the Kármán vortex street behind cylinders and inclined flat plates see [46, pp. 56–57]. The level of detail is quite remarkable, and provides a visual impression of the flow that is superior to any Eulerian criterion used to extract vortical structures [29]. The associated mesh clearly shows that nodes are mainly located on the *LCS*, and provides a skeleton representation of the vortical structures. While examining the flow shows this 2D cut to be quite well organized, results shown in the symmetry plane parallel to the leaflet in Fig. 10(b) reveal that strong perturbations develop inside the jet that forms between the two leaflets. This is apparent from the FTLE contours and from the mesh, which shows the high density of nodes in this plane.

The level of grid refinement required by the analysis of the perturbed flow that forms close to the pipe axis strongly contrasts with that of the more quiescent flow that develops around the leaflets in the vicinity of the pipe wall. Therefore, this example illustrates the advantage of using mesh adaptation in flows where boundaries exist between laminar and turbulent regions, characterized by velocity scales separated by several orders of magnitude. This includes most flows studied in indus-

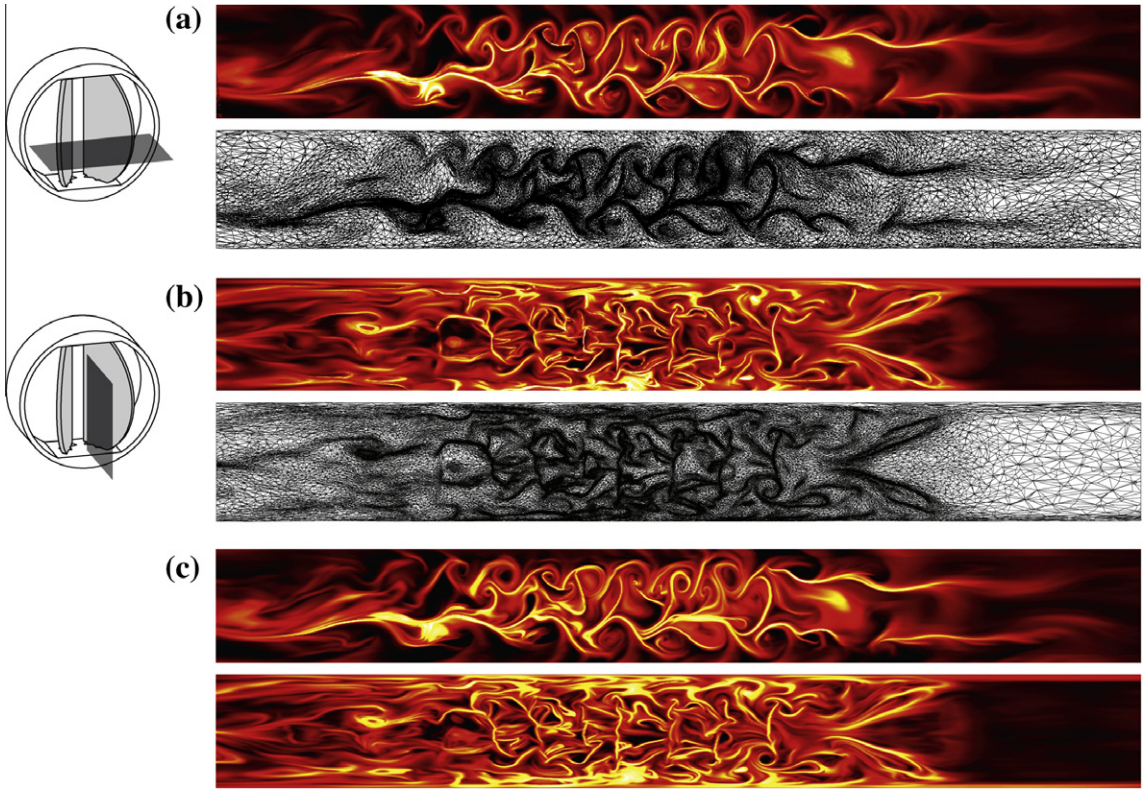


Fig. 10. 3D FTLE field computed downstream of a bileaflet mechanical heart valve with mesh adaptation. The cut-plane of the FTLE field is indicated in the geometry of the valve shown on the left. The resulting contour levels and meshes are shown in (a) the symmetry plane perpendicular to the leaflets and (b) the symmetry plane parallel to the leaflets. Results obtained without mesh adaptation are shown in (c) for comparison.

trial and research applications, such as shear layers, jets, and wakes. In order to give an idea of the benefit of mesh adaptation, Fig. 10(c) shows the results obtained using a Cartesian grid. The initial grid has been refined until a result is obtained that compares to that obtained with the mesh adaptation. Although the level of detail is not as high as in Fig. 10(a) and (b) (in particular the sharpness of the vortical structures), the Cartesian grid used 34 million nodes, compared with 1.7 million for the anisotropic mesh, which gives a ratio of 20:1. The size of the ASCII data file yielding these results is 400 MB with mesh adaptation and 2.3 GB without. The gain obtained in the file size is smaller than the gain obtained by the reduction of nodes since unstructured meshes require the node connectivity.

8. Conclusion

The identification of flow structures by the finite-time Lyapunov exponent (FTLE) has been revisited, both to enhance accuracy and to simultaneously adapt the number of nodes in the computational grid. The methodology, based on a set of 12 differential equations in 2D and 30 differential equations in 3D, yields the flow map and its first and second derivatives, and intrinsically provides an interpolation error as a quality indicator of the FTLE field. This results in improved nodal accuracy of the components of the Cauchy-Green tensor and its eigenvalues. With the Hessian of the flow map, we define, in 3D, three metrics that intersect in a single, continuous, and optimal multiscale metric. This metric is read into anisotropic mesh adaptation software to iteratively reveal the flow structures based on an appropriate L^p metric.

The efficiency of the method is illustrated with several examples. Results obtained from analytical solutions, some of which have already been used in previous works, reveal that the current framework of anisotropic mesh adaptation is well suited to the computation of FTLE fields, with an element aspect ratio reaching 1/1,000,000. This is confirmed in experimental test cases, where both laminar and turbulent flows are particularly well detailed and for which the interpolation error estimation gives a quadratic convergence, as theoretically predicted. While the use of the L^∞ metric best highlights Lagrangian Coherent Structures in laminar flows by minimizing the number of nodes, less well defined structures are best revealed by the use of the L^1 metric in turbulent flows. In practice, both metrics converge to the same results if the number of nodes is sufficiently high. In particular, this number of nodes is at least one order of magnitude lower than that required with a regular Cartesian grid.

Although the computational method is much more efficient in capturing *LCS* structures than with successively refined Cartesian grids, the final anisotropic mesh is generally obtained by gradually increasing the mesh complexity (number of nodes) and the integration time of particle trajectories to avoid premature breakdown of the mesh adaptation software. For inexperienced users, this can be particularly time consuming. For the most demanding applications, such as 3D turbulent flows, which will probably be investigated in the future owing to increasingly capable velocity measurement techniques, mesh adaptation would likely need to be combined with fast FTLE field computation methods, which is currently an active area of research.

The following guidelines are indented as they are recommendations for increasing the robustness of the anisotropic mesh adaptation along *LCS* structures. The adaptation process begins with a small integration time T to smooth out the *LCS* structures. This simplifies the first adaptation steps, since the mesh usually goes through very significant deformation along the complex ridge topology. Once the topology has been captured, the integration time is slowly increased incrementally to avoid any abrupt changes in the topology of the structures that could lead to breakdown of the adaptation. This sharpens the ridges and increases the mesh density and element stretching along these structures. The maximum allowed integration time T is, however, limited by the heuristics built into the adaptation software to manage the rapid transition between nearly isotropic mesh regions to highly anisotropic mesh regions.

The methodology that we have described in this paper is currently restricted to linear finite element interpolation of the FTLE field, and is well suited to Hessian-based anisotropic mesh adaptation. However, the quality of the FTLE field representation could also be improved with higher order polynomial approximation, and is certainly a path to follow to reduce the number of nodes and the anisotropic ratio near *LCS*. This would certainly benefit mesh adaptation software robustness and efficiency. In theory, it should have no impact on the computation of the flow map and its gradient, and should improve FTLE field quality. However, the generalization of our approach to P^k polynomials would require an excessive numbers of coupled differential equations. This suggests the need to develop a different anisotropic mesh adaptation framework, as indicated by recent advancements in this field of research [47].

Acknowledgements

The authors would like to acknowledge the financial support of the Natural Sciences and Engineering Research Council of Canada and of the Canada Foundation for Innovation.

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